Data Analytics & Operations Improvement

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1.1 Project Context

The case studies were developed from real-world operational improvement work at a gold-producing mining operation. The projects demonstrate the application of data analytics, KPI modelling, root cause analysis, business case development, and benefit tracking to inform operational decisions and drive continuous improvement.
Operational improvements were implemented through collaboration among Operations, Maintenance, Engineering, Technical Services, Finance, and HSE.

- What This Portfolio Is

Six operational improvement initiatives developed from mine control data, structured into business cases, implemented through cross-functional collaboration, and tracked to verified financial outcomes.

- Portfolio Summary
  - 6 operational improvement initiatives
  - $4.97M annualized business value
  - $0 capital expenditure required
  - 36% average KPI improvement

1.2 Methodology

1.3 Portfolio Projects- The Six Initiatives

1. *Diamond Drill Self-Perform* - Cost Optimization| *$1.8M annual savings*

Drilling cost per meter dropped 52% from $272.60 to $130.30, delivering $1.8M in annual savings- by building a contractor-vs-self-perform cost model from monthly drilling and finance records and creating the first formal benefit-tracking system for the newly established in-house drilling team. Total monthly drilling cost decreased from $140,928 to $69,758, a 50.5% reduction.

2. Underground Tyre Damage Reduction - *Equipment & Fleet | $1.04M annual savings*

Across a fleet of 21 trucks, each with 12 tyres, haul truck tyre failures fell 26%, from 7.8 tyres/day to 5.8 tyres/ day, saving $1.04M annually. This was achieved by conducting a root cause analysis of daily mine-control tyre downtime and maintenance-related tire failure data. Identified road surface degradation, spill mismanagement, and differential lock overuse as the primary failure drivers. By coordinating road conditioning, revised spill procedures, and operator retraining with the Underground Operations and Maintenance teams. Average repair duration per failure also fell 26.7%, from 4.1 to 3.0 hours.

3. Open Pit Tyre Life Improvement - Equipment & Fleet | *$723K annual savings*

Across a 19-truck Volvo FMX 480 haul fleet, open-pit tyre consumption dropped 33% - from 62.9 to 42.0 tyres per month, saving $723K annually, by applying Pareto analysis to monthly tyre procurement and downtime records to isolate the highest-cost failure types, then structuring a haul road maintenance programme and operator training plan with the Open Pit Operations team. Monthly tyre cost fell from $56,768 to $37,931, a matching 33.2% reduction.

4. Jumbo Drilling Cycle Reduction - Underground Development Productivity | *$704K annual savings*

Underground jumbo development rate increased by 63%, from 3.2 to 5.2 meters per day, by reducing cutting cycle time from 32.1 to 27.0 hours per 3m cut and delivering $704K annually. Decomposed the full development cycle into timed subtasks, benchmarked each against experienced-operator standards, and coordinated shift readiness protocols, targeted training, and logistics adjustments with the Supply chain and Maintenance teams to eliminate waste.

5. Underground Development Overbreak Reduction - Underground Development Productivity | $492K annual savings

Development overbreak fell 22% (from 12.8% to 10.0%), saving $492K annually in loading and haulage costs. This was achieved by applying fishbone analysis to monthly development data to identify drill hole parallelism, face markup accuracy, and charge density as controllable root causes, and by coordinating face-marking training, blasting audits, and controlled blasting techniques with the Technical Services and HSE teams.

6. LHD Loader Utilization - Production & Development Productivity | $217K annual savings

Loader availability increased by 8.1 percentage points, from 52.6% to 60.7%, and utilization rose by 3.3 percentage points, from 68.8% to 72.1%. The improvement was achieved by analysing shift logs, equipment performance data, and maintenance data to identify gaps in real-time reporting. Finally, live mine-control data capture was implemented, operators were deployed for hot-seating, and an underground service bay was established to reduce breakdown turnaround time.

1.4 Skills Demonstrated

Excel (PivotTables, time-series analysis, KPI Modelling, Dashboards)
Data Analysis & Visualization, Business Case Quantification
Value Driver Trees (VDTs)
Root Cause Analysis, Fishbone diagrams, Pareto analysis, 5S, Statistics
Lean Six Sigma / Continuous Improvement
Process Mapping
Stakeholder Management & Engagement
Benefit Tracking
Operational Analytics, Control Room Data Interpretation

1.5 About the Data

This workbook uses an anonymized case study ("Goldridge Mining Operations"). All company names, personnel, and site identifiers are fictional. The datasets, KPIs, and workflows reflect realistic mining operational patterns and the methodology used in a real production environment, while protecting confidential business information. In a live setting, source data flows directly from the mine control system.

1.6 Portfolio Purpose

This portfolio shows how operational data can be transformed into measurable business outcomes through structured analysis, root cause investigation, stakeholder engagement, implementation support, and benefit tracking. It focuses on the analytical process of identifying opportunities, prioritizing actions, quantifying value, measuring performance, sustaining performance, and supporting operational improvements.

Portfolio Note:
This work is intended to demonstrate analytical thinking, operational problem-solving, and business value realisation using anonymised operational datasets and representative mining scenarios.